



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 38

Level 2 — Intermediate · Data Handling

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Data Handling — Scaled Graphs & Tables
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	38 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 38 levels up data reading with SCALES: a pictograph where one circle = 5 cars, a bar graph in hundreds, 'how many times' comparisons ($2.5\times$), marks LOST out of a total, and combined questions that mix the chart with a calculation (trees $\times$ buckets of water). Reading the scale first is the habit this sheet drills.

Learning Objectives

- Multiply by the scale when reading pictographs and bar graphs.
- Compute totals, differences and 'how many times' from graphs.
- Work with marks lost vs marks scored from a data table.
- Combine chart data with a further calculation step.
- Compare tally-chart totals with bundles of five.

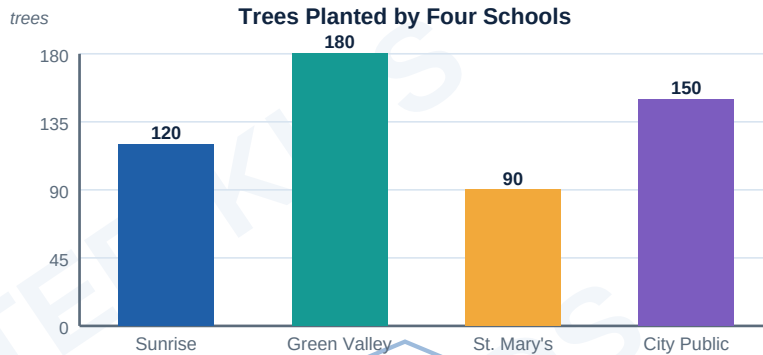
Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

Q1. The bar graph shows trees planted by four schools. Which school planted the MOST trees?



- (A) Sunrise
(B) Green Valley
(C) St. Mary's
(D) City Public

Q2. (Same tree graph.) How many trees did the four schools plant in ALL?

- (A) 520
(B) 540
(C) 560
(D) 500

Q3. (Same tree graph.) How many MORE trees did Green Valley plant than St. Mary's?

- (A) 60
(B) 70
(C) 80
(D) 90

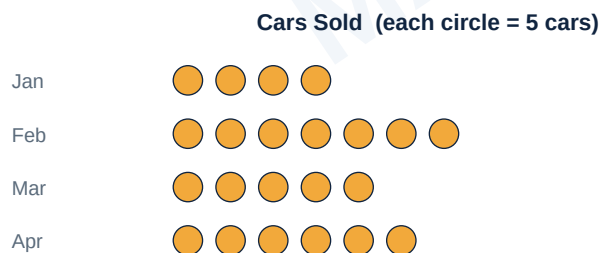
Q4. (Same tree graph.) Which TWO schools together planted exactly 300 trees?

- (A) Sunrise and Green Valley
(B) Sunrise and City Public
(C) Green Valley and St. Mary's
(D) St. Mary's and City Public

Q5. (Same tree graph.) If each tree needs 2 buckets of water a week, how many buckets does City Public's plantation need weekly?

- (A) 150
(B) 300
(C) 200
(D) 250

Q6. The pictograph shows cars sold (each circle = 5 cars). How many cars were sold in FEBRUARY?



- (A) 7
(B) 25
(C) 35
(D) 30

Q7. (Same pictograph.) How many cars were sold in January and March TOGETHER?

- (A) 9 (B) 45
(C) 40 (D) 35

Q8. (Same pictograph.) In which month were exactly 30 cars sold?

- (A) January (B) February
(C) March (D) April

Q9. (Same pictograph.) How many MORE cars were sold in February than in January?

- (A) 3 (B) 15
(C) 20 (D) 10

Q10. Aditi's test marks (out of 50): Maths 42, Science 38, English 45, Hindi 35. In which subject did she score the HIGHEST?

- (A) Maths (B) Science
(C) English (D) Hindi

Q11. (Same marks.) What is Aditi's TOTAL score across the four subjects?

- (A) 150 (B) 160
(C) 170 (D) 165

Q12. (Same marks: Maths 42, Science 38, English 45, Hindi 35, each out of 50.) How many marks did Aditi LOSE in all (out of the total 200)?

- (A) 30 (B) 40
(C) 50 (D) 45

Q13. (Same marks.) In which subject did she lose the MOST marks?

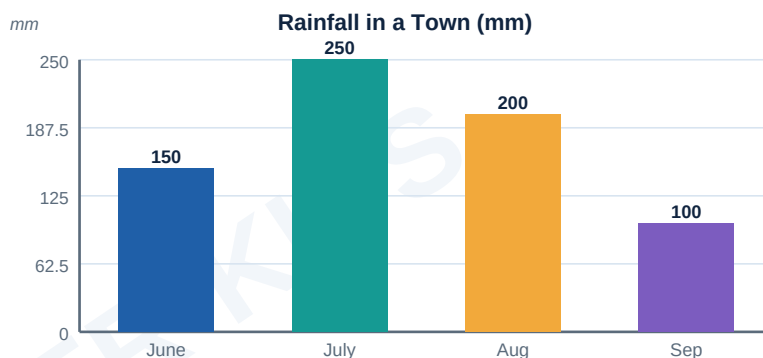
- (A) Maths (B) Science
(C) English (D) Hindi

Q14. (Same marks.) The difference between her BEST and WORST scores is:

- (A) 5 (B) 10
(C) 15 (D) 20



Q15. The graph shows monthly rainfall. The TOTAL rainfall over the four months is:



(A) 600 mm

(C) 700 mm

(B) 650 mm

(D) 750 mm

Q16. (Same rainfall graph.) July's rainfall is how many TIMES September's?

(A) 2 times

(C) 3 times

(B) 2.5 times

(D) 1.5 times

Q17. (Same rainfall graph.) Arrange the months from WETTEST to DRIEST.

(A) July, Aug, June, Sep

(C) Sep, June, Aug, July

(B) July, June, Aug, Sep

(D) Aug, July, June, Sep

Q18. In a pictograph, HALF a symbol is sometimes drawn. If one symbol = 10 books, half a symbol means:

(A) 10 books

(C) 2 books

(B) 5 books

(D) 20 books

Q19. A tally chart shows: Cricket — 4 crossed bundles and 2 singles; Football — 3 crossed bundles and 4 singles. How many MORE children chose cricket?

(A) 3

(C) 5

(B) 4

(D) 6

Q20. A bar graph's scale shows 1 unit = 20 students. A bar that is 6 units tall stands for:

(A) 6 students

(C) 120 students

(B) 26 students

(D) 200 students



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	D	A	B	C	B	D	B	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	B	D	B	C	B	A	B	A	C

Step-by-Step Solutions

Q1. Correct answer: (B)

The tallest bar is **Green Valley** with 180 trees.

Q2. Correct answer: (B)

$120 + 180 + 90 + 150 = 540$ trees.



Q3. Correct answer: (D)

$180 - 90 = 90$ trees. (Green Valley planted exactly DOUBLE St. Mary's!)

Q4. Correct answer: (A)

Sunrise + Green Valley = $120 + 180 = 300$. (Others: 270, 270, 240.)

Q5. Correct answer: (B)

$150 \text{ trees} \times 2 \text{ buckets} = 300$ buckets.

Q6. Correct answer: (C)

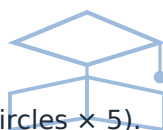
February shows $7 \text{ circles} \times 5 = 35$ cars. Always multiply by the scale!

Q7. Correct answer: (B)

Jan = $4 \times 5 = 20$; Mar = $5 \times 5 = 25$. Together = **45 cars**.

Q8. Correct answer: (D)

$30 \div 5 = 6$ circles → **April**.



Q9. Correct answer: (B)

Feb – Jan = $35 - 20 = 15$ cars (or 3 extra circles $\times 5$).

Q10. Correct answer: (C)

Compare: 42, 38, **45**, 35. Highest is English.

Q11. Correct answer: (B)

$42 + 38 + 45 + 35 = 160$.

Q12. Correct answer: (B)

Total possible = 200; scored = 160. Lost = $200 - 160 = 40$ marks.

Q13. Correct answer: (D)

Lost: Maths 8, Science 12, English 5, **Hindi 15**. Most lost in Hindi (lowest score = most lost).

Q14. Correct answer: (B)

Best 45 – worst 35 = **10 marks**.

Q15. Correct answer: (C)

$150 + 250 + 200 + 100 = 700$ mm.



Q16. Correct answer: (B)

$250 \div 100 = 2.5$ times (two and a half times).

Q17. Correct answer: (A)

$250 > 200 > 150 > 100 \rightarrow$ **July, Aug, June, Sep.**

Q18. Correct answer: (B)

Half of the scale: $10 \div 2 = 5$ books.

Q19. Correct answer: (A)

Cricket = $4 \times 5 + 2 = 22$; Football = $3 \times 5 + 4 = 19$. Difference = **3**.

Q20. Correct answer: (C)

$6 \times 20 = 120$ students. Read the SCALE before reading the bars.

